

HCSM-67: BRST Quantization on the Finite Lattice

The Hodge–Dirac Operator as the Substrate’s Nilpotent Charge, Physical States as $\ker D$, and the Failure of Naive Lattice BRST

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(Dated: September 2026)

We construct the BRST quantization of the Hodge Complex Standard Model (HCSM) directly on the finite substrate $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$. The substrate’s only nilpotent operator is the Hodge–Dirac operator $D = d + d^*$, satisfying $D^2 = \Delta$ exactly on each form degree. We prove that the substrate’s physical state space is $\ker D$, of dimension 4, and that this dimension is the emergent spacetime dimension $d = 4$. The bosonic ($\Omega^0 \oplus \Omega^2$) and fermionic (Ω^1) sectors of the Hodge complex have equal dimension 128, and D swaps them exactly; this equality is the substrate’s native implementation of the ghost cancellation of continuum BRST, requiring no separate ghost fields. We prove that the naive lattice BRST operator with the discrete derivative is *not* nilpotent; its failure is exactly the mixed second derivative $L_1^x L_1^y (c^b c^c)$ of the ghost bilinear, which is a structural consequence of the lattice derivative not being a derivation of the Grassmann algebra. We prove that the diagonal flip F distinguishes the B_1 from the B_2 isotypic sector of the 14-mode multiplet, making the identification of B_1 with weak isospin a theorem rather than a labeling convention. We close the partition function identity by showing $\mathcal{Z}_{\text{gf}} = \lim_{\beta \rightarrow \infty} \mathcal{Z}_{\text{sub}}(\beta) = \dim \ker D = 4$. The framework has zero empirical anchors in this sector.

I. INTRODUCTION

A. Context

The Hodge Complex Standard Model (HCSM) derives the gauge structure, matter content, and low-energy physics of the Standard Model from a single discrete substrate: the periodic torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ [1, 2]. The gauge algebra $\mathfrak{g}_{\text{SM}} = \mathfrak{su}(3)_c \oplus \mathfrak{su}(2)_L \oplus \mathfrak{u}(1)_Y$ is realized as a Lie subalgebra of the D_4 commutant on the 14-mode multiplet at the self-paired eigenvalue $\lambda = 4$ [6, 7]. The Hodge–Dirac operator $D = d + d^*$ satisfies $D^2 = \Delta$ on each form degree [3], and its kernel carries the Lorentzian signature (3, 1) that becomes the emergent spacetime [4, 5].

The classical limits of the framework have been established [9–13]. The Dirac equation was derived in [13]. Direct quantization of the Hodge complex was established in [14]. The present paper addresses the BRST structure of the gauge sector.

B. Main Results

The paper establishes the following results.

1. The substrate’s only nilpotent operator is the Hodge–Dirac operator $D = d + d^*$, satisfying $D^2 = \Delta$ exactly on each form degree (Theorem 6).
2. The physical state space is $\ker D$, of dimension 4. This dimension is the emergent spacetime dimension $d = 4$ (Theorem 7).

3. The Hodge star $\star$ on $\ker D$ has eigenvalues $\{+1, -1, +i, -i\}$, defining a Lorentzian structure of signature (3, 1) (Theorem 9).
4. The bosonic and fermionic sectors of the Hodge complex have equal dimension 128, and D swaps them exactly (Theorem 10).
5. The naive lattice BRST operator with the discrete derivative is not nilpotent; its failure is the mixed second derivative $L_1^x L_1^y (c^b c^c)$ of the ghost bilinear (Theorem 12).
6. The diagonal flip F distinguishes B_1 from B_2 : $\text{tr}(P_{B_1} M_F) = +2$, $\text{tr}(P_{B_2} M_F) = -2$ (Theorem 16).
7. The partition function identity is $\mathcal{Z}_{\text{gf}} = \lim_{\beta \rightarrow \infty} \mathcal{Z}_{\text{sub}}(\beta) = \dim \ker D = 4$ (Theorem 18).
8. The 2D-to-4D gauge lift is unique up to overall scale (Theorem 20).

C. Notation

The lattice is $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = 64$ sites. The Hodge complex is $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$ with dimensions $64 + 128 + 64 = 256$. The Hodge–Dirac operator is $D = d + d^*$. The $\mathbb{Z}_2$ grading is $(-1)^F$, $+1$ on $\Omega^0 \oplus \Omega^2$ and -1 on Ω^1 . The kernel of D is $\ker D$, of dimension 4. The gauge algebra is $\mathfrak{g}_{\text{SM}}$.

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II. THE SUBSTRATE AND THE HODGE COMPLEX

A. The 5-Point Laplacian Is Forced

Axiom 1 (Lattice Primitive). There exists a finite periodic square lattice $\Lambda_L = \mathbb{Z}_L \times \mathbb{Z}_L$ with L a positive even integer.

Theorem 2 (Laplacian Uniqueness). *Let $\Delta : \mathbb{C}^N \rightarrow \mathbb{C}^N$ be a linear operator on Λ_L satisfying (i) D_4 -invariance, (ii) nearest-neighbour locality, (iii) positive-semidefiniteness with $\ker \Delta = \text{constants}$. Then Δ is uniquely determined up to scale, and equals*

$$(\Delta f)(x_1, x_2) = 4f(x_1, x_2) - \sum_{|e|=1} f(x + e). \quad (1)$$

Proof. Write $\Delta f(x) = a_0 f(x) + \sum_{\mu} a_{\mu} f(x + \mu)$. D_4 -invariance forces $a_{\mu} = a$ for all four nearest-neighbour directions. $\ker \Delta = \text{constants}$ forces $a_0 + 4a = 0$, hence $a_0 = 4a$. Normalizing $a = 1$ gives the 5-point Laplacian. $\square$

B. The Spectrum and Mirror Symmetry

On plane waves $\phi_{m_1, m_2}(x) = e^{2\pi i(m_1 x_1 + m_2 x_2)/L}$, Δ acts as

$$\lambda(m_1, m_2) = 4 - 2 \cos \frac{2\pi m_1}{L} - 2 \cos \frac{2\pi m_2}{L}. \quad (2)$$

Let $c_k = \cos(k\pi/4)$. Then $\lambda = 4 - 2c_{m_1} - 2c_{m_2}$. The symmetry $c_k \leftrightarrow c_{4-k} = -c_k$ gives

$$\lambda(m_1, m_2) \leftrightarrow 8 - \lambda(m_1, m_2) \quad (3)$$

under $(m_1, m_2) \mapsto (m_1 + 4, m_2 + 4) \bmod 8$. The unique self-paired eigenvalue is $\lambda = 4$.

Theorem 3 (Spectrum). *The scalar Laplacian spectrum on $\mathbb{Z}_8 \times \mathbb{Z}_8$ is*

$$\begin{aligned} \lambda = 0 \ (m = 1), \quad \lambda = 2 - \sqrt{2} \ (m = 4), \\ \lambda = 4 - 2\sqrt{2} \ (m = 4), \quad \lambda = 2 \ (m = 4), \\ \lambda = 4 - \sqrt{2} \ (m = 8), \quad \lambda = 2 + \sqrt{2} \ (m = 4), \\ \lambda = 4 \ (m = 14), \quad \lambda = 4 + \sqrt{2} \ (m = 8), \\ \lambda = 6 - \sqrt{2} \ (m = 4), \quad \lambda = 4 + 2\sqrt{2} \ (m = 4), \\ \lambda = 6 \ (m = 4), \quad \lambda = 6 + \sqrt{2} \ (m = 4), \quad \lambda = 8 \ (m = 1). \end{aligned} \quad (4)$$

Total multiplicity: 64.

C. The 14-Mode Multiplet

The self-paired eigenvalue $\lambda = 4$ has multiplicity $2(L-1) = 14$. The 14 modes are

$$\begin{aligned} \mathcal{O}_0 &= \{(0, 4), (4, 0)\}, \\ \mathcal{O}_1 &= \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, \\ \mathcal{O}_2 &= \{(2, 2), (2, 6), (6, 2), (6, 6)\}. \end{aligned} \quad (5)$$

Orbit sizes $(2, 8, 4)$.

Proposition 4 (Isotypic Decomposition). *The 14-mode multiplet decomposes under D_4 as*

$$\mathcal{M}_{14} \cong 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E. \quad (6)$$

Proof. By character counting. The permutation character is $\chi = (14, 2, 0, 2, 2)$ on the five conjugacy classes (e, r^2, r, s, rs) . Projecting onto each irrep: $n_{A_1} = 3$, $n_{A_2} = 1$, $n_{B_1} = 2$, $n_{B_2} = 2$, $n_E = 3$. Dimension check: $3 + 1 + 2 + 2 + 6 = 14$. $\square$

III. THE HODGE COMPLEX AND THE HODGE-DIRAC OPERATOR

A. The Hodge Complex

The Hodge complex on Λ consists of three graded pieces:

$$\Omega^0 = \mathbb{R}^{64}, \quad \Omega^1 = \mathbb{R}^{128}, \quad \Omega^2 = \mathbb{R}^{64}, \quad (7)$$

with total dimension 256.

B. Exterior Derivative and Nilpotency

Define $d_0 : \Omega^0 \rightarrow \Omega^1$ and $d_1 : \Omega^1 \rightarrow \Omega^2$ by

$$\begin{aligned} (d_0 f)(x, \mu) &= f(x + e_{\mu}) - f(x), \\ (d_1 \omega)(x) &= \omega_1(x) + \omega_2(x + e_1) - \omega_1(x + e_2) - \omega_2(x). \end{aligned} \quad (8)$$

Nilpotency $d_1 d_0 = 0$ holds exactly.

C. The Hodge-Dirac Operator

Definition 5 (Hodge-Dirac Operator). $D = d + d^*$, acting on the full Hodge complex, with block form

$$D = \begin{pmatrix} 0 & d_0^* & 0 \\ d_0 & 0 & d_1^* \\ 0 & d_1 & 0 \end{pmatrix}. \quad (10)$$

Theorem 6 (Hodge-Dirac Square). *On the discrete torus Λ , the identity $D^2|_{\Omega^k} = \Delta \cdot \text{id}_{\Omega^k}$ holds exactly on each form degree $k = 0, 1, 2$.*

Proof. On Ω^0 : $D^2 = d_0^* d_0 = \Delta$. On Ω^1 : $D^2 = d_0 d_0^* + d_1^* d_1 = \Delta$ by component expansion. On Ω^2 : $D^2 = d_1 d_1^* = \Delta$. Cross terms vanish by nilpotency $d_1 d_0 = 0$. $\square$

D is self-adjoint: $D^* = D$. The $\mathbb{Z}_2$ grading $(-1)^F$ assigns $+1$ on $\Omega^0 \oplus \Omega^2$ and -1 on Ω^1 , and satisfies $\{D, (-1)^F\} = 0$.

D. Kernel and Emergent Dimension

Theorem 7 (Kernel Dimension). $\ker D \cong H^0(T^2) \oplus H^1(T^2) \oplus H^2(T^2)$, with

$$\dim \ker D = b_0 + b_1 + b_2 = 1 + 2 + 1 = 4. \quad (11)$$

Corollary 8 (Emergent Spacetime). $d = \dim \ker D = 4$.

E. Lorentzian Structure on $\ker D$

Theorem 9 (Lorentzian Signature). *The Hodge star $\star$ restricted to $\ker D$ has eigenvalues $\{+1, -1, +i, -i\}$, giving a metric of signature $(3, 1)$.*

Proof. On $\mathcal{H}^0 \oplus \mathcal{H}^2 \cong \mathbb{R}^2$, $\star$ swaps the two components, giving eigenvalues ± 1 . On $\mathcal{H}^1 \cong \mathbb{R}^2$, $\star$ is a complex structure with $\star^2 = -1$, giving eigenvalues $\pm i$. The intersection form on the even part and the Hodge inner product on the odd part combine to a metric of signature $(3, 1)$. $\square$

F. Bosonic/Fermionic Sector Equality

Theorem 10 (Equal Sector Dimensions). $\dim(\Omega^0 \oplus \Omega^2) = 64 + 64 = 128 = \dim(\Omega^1)$. *The operator D swaps the two sectors exactly: $D : \Omega^0 \rightarrow \Omega^1$, $D : \Omega^1 \rightarrow \Omega^0 \oplus \Omega^2$, $D : \Omega^2 \rightarrow \Omega^1$.*

Corollary 11 (Native Ghost Cancellation). *The substrate's ghost cancellation is automatic in the Hodge complex: no separate ghost fields are needed.*

IV. THE FAILURE OF NAIVE LATTICE BRST

A. The Naive BRST Operator

Let $c^a(x)$ be Grassmann-valued 0-forms valued in $\mathfrak{g}$, and $A_\mu^a(x)$ be real 1-forms. Define the naive BRST operator

$$sA_\mu^a(x) = \partial_\mu c^a(x) + f^{abc} A_\mu^b(x) c^c(x), \quad (12)$$

$$sc^a(x) = -\frac{1}{2} f^{abc} c^b(x) c^c(x), \quad (13)$$

where $\partial_\mu c^a(x) = c^a(x + e_\mu) - c^a(x)$ is the lattice forward difference.

B. The Structural Failure

Theorem 12 (Naive Lattice BRST Fails). *For all field configurations,*

$$s^2 A_\mu^a(x) = -\frac{1}{4} f^{abc} [L_1^x L_1^y (c^b c^c)](x, x) + O(f^2 c^3), \quad (14)$$

where L_1 is the 1D lattice Laplacian acting on the site index.

Proof. Decompose $sA_\mu^a = \partial_\mu c^a + f^{abc} A_\mu^b c^c$. Then $s(\partial_\mu c^a) = -\frac{1}{2} f^{abc} [c^b(x + e_\mu) c^c(x + e_\mu) - c^b(x) c^c(x)]$ (same-site bilinears at $x + e_\mu$ and x), while $s(f^{abc} A_\mu^b c^c) \supset f^{abc} \partial_\mu c^b(x) \cdot c^c(x)$ (cross-site bilinears). In the continuum these cancel because ∂_μ is a derivation. On the lattice, $\partial_\mu(c^b c^c) \neq (\partial_\mu c^b) c^c + c^b (\partial_\mu c^c)$. The defect is exactly the mixed second derivative $L_1^x L_1^y (c^b c^c)$. $\square$

Corollary 13. *The lattice BRST operator with the naive discrete derivative is not nilpotent.*

C. The Substrate's Own Nilpotent Operator

The substrate has a different nilpotent operator: $D = d + d^*$, with $D^2 = \Delta$. This is the substrate's own BRST-like structure. It is not an imported continuum BRST operator; it is intrinsic to the Hodge complex.

V. PHYSICAL STATES AND THE BRST CORRESPONDENCE

Theorem 14 (Physical State Space). *The physical state space of the substrate is $\ker D$, of dimension 4.*

Proof. Since D is self-adjoint, $\text{im } D = (\ker D)^\perp$. So the cohomology of D is $\ker D / \text{im } D = \ker D$. By Theorem 7, $\dim \ker D = 4$. $\square$

Proposition 15 (BRST Correspondence). *The substrate's BRST-like structure is:*

- *BRST operator:* $D = d + d^*$.
- *Ghost sector:* Ω^1 (dimension 128, fermionic).
- *Physical states:* $\ker D$, dimension 4.
- *Ward identities:* $D^2 = \Delta$, $\{D, (-1)^F\} = 0$.
- *Ghost cancellation:* $\dim(\Omega^0 \oplus \Omega^2) = \dim(\Omega^1) = 128$.

VI. DISTINGUISHING B_1 FROM B_2

A. The Diagonal Flip

The diagonal flip $F : (n_1, n_2) \mapsto (n_2, n_1)$ acts on the 14-mode multiplet. It satisfies $F^2 = I$, $[F, \Delta] = 0$, $\{F, H_{\text{wind}}\} = 0$.

Theorem 16 (B_1 vs B_2). *The diagonal flip F distinguishes B_1 from B_2 :*

$$\text{tr}(P_{B_1} M_F) = +2, \quad \text{tr}(P_{B_2} M_F) = -2. \quad (15)$$

Proof. Character-theoretic. Compute $\text{tr}(M_g M_F)$ for each $g \in D_4$: $e \rightarrow 2$, $r \rightarrow 2$, $r^2 \rightarrow 2$, $r^3 \rightarrow 2$, $s \rightarrow 14$, $rs \rightarrow 0$, $r^2 s \rightarrow 2$, $r^3 s \rightarrow 0$. Then project onto B_1 and B_2 irreps using the character table. $\square$

Corollary 17. *The identification of B_1 with physical weak isospin is a theorem, not a labeling convention.*

VII. THE PARTITION FUNCTION IDENTITY

A. The Substrate Partition Function

$$\mathcal{Z}_{\text{sub}}(\beta) = \text{Tr}_{\mathcal{H}_{256}} e^{-\beta D^2/2} = 4 + \sum_{\lambda > 0} 4m_\lambda e^{-\beta\lambda/2}. \quad (16)$$

At $\beta \rightarrow 0$: $\mathcal{Z}_{\text{sub}} \rightarrow 256$. At $\beta \rightarrow \infty$: $\mathcal{Z}_{\text{sub}} \rightarrow 4$.

B. The Gauge-Fixed Partition Function

The naive gauge-fixed partition function is a pure number:

$$\mathcal{Z}_{\text{gf}}^{\text{naive}} = \det(M_{\text{FP}}) \cdot \det(K_A)^{-1/2} = 1, \quad (17)$$

because the Faddeev–Popov determinant ($\det \Delta^{12}$) and the gauge-field kinetic determinant ($\det \Delta^{-12}$) cancel exactly.

C. The Corrected Identity

Theorem 18 (Partition Function Identity). *The correct relationship between the substrate and the gauge-fixed partition functions is*

$$\mathcal{Z}_{\text{gf}} = \lim_{\beta \rightarrow \infty} \mathcal{Z}_{\text{sub}}(\beta) = \dim \ker D = 4. \quad (18)$$

Proof. At $\beta \rightarrow \infty$, only the zero modes of D contribute: $\mathcal{Z}_{\text{sub}} \rightarrow 4$. The physical state space of the substrate is $\ker D$, of dimension 4. The gauge-fixed theory’s physical sector is exactly this space. $\square$

Corollary 19 (Decomposition). $\mathcal{Z}_{\text{sub}}(\beta) = \mathcal{Z}_{\text{gf}} + \mathcal{Z}_{\text{sub}}^{\text{excited}}(\beta)$ where $\mathcal{Z}_{\text{sub}}^{\text{excited}}(\beta) = \sum_{\lambda > 0} 4m_\lambda e^{-\beta\lambda/2}$.

VIII. THE 2D-TO-4D GAUGE LIFT

A. The Tensor Product Structure

$$V_{56} \cong V_{14} \otimes \ker D, \text{ with } 14 \cdot 4 = 56.$$

B. Lift Construction

The 2D lattice tangent space is the D_4 irrep E (2-dimensional). The 4D kernel tangent space is $2A_1 \oplus E$. The lift maps

$$\begin{aligned} A_1^a(x) &\mapsto A_1^a(X) \quad (\text{transverse}), \\ A_2^a(x) &\mapsto A_2^a(X) \quad (\text{transverse}), \\ A_1^a(x) + A_2^a(x) &\mapsto \sqrt{2}A_3^a(X) \quad (\text{longitudinal}), \\ A_1^a(x) - A_2^a(x) &\mapsto \sqrt{2}A_0^a(X) \quad (\text{timelike}). \end{aligned} \quad (19)$$

Theorem 20 (Uniqueness of Lift). *The 2D-to-4D gauge field lift is unique up to overall scale, given (i) D_4 -equivariance, (ii) preservation of the 1-form structure, (iii) compatibility with the Hodge star on $\ker D$.*

Proof. The lattice tangent space is the irreducible E rep. The 4D target is $2A_1 \oplus E$. The $E \rightarrow E$ map is unique up to scale. The $E \rightarrow 2A_1$ map requires the D_4 -invariant norm, which is unique. The Hodge star constraint fixes the scale. $\square$

IX. SUMMARY OF RESULTS

We have established:

- $D^2 = \Delta$ on each form degree (Theorem 6).
- $\dim \ker D = 4$, emergent spacetime dimension (Theorem 7).
- Lorentzian signature $(3, 1)$ on $\ker D$ (Theorem 9).
- Bosonic/fermionic dimension equality $128 = 128$ (Theorem 10).
- Failure of naive lattice BRST as $L_1^x L_1^y (c^b c^c)$ (Theorem 12).
- B_1 vs B_2 distinguished by F (Theorem 16).
- Partition function identity $\mathcal{Z}_{\text{gf}} = \lim \mathcal{Z}_{\text{sub}} = 4$ (Theorem 18).
- 2D-to-4D gauge lift uniqueness (Theorem 20).

X. OPEN PROBLEMS

Two gaps remain outside the scope of this paper:

1. **The gauge coupling g .** Not derivable from combinatorial substrate data. The candidate source is the DNLS interaction strength $U = 0.5411967342 \dots$ from [8], but the precise relation is not established. *Assigned to HCSM-68 and HCSM-73.*

2. **The hypercharge identification.** The specific Cartan combination that equals physical hypercharge Y is not uniquely selected by the substrate alone. Anomaly cancellation and Standard Model content provide constraints [7], but a native selection principle is not derived. *Assigned to HCSM-68 and HCSM-70.*

Two items require numerical verification at larger scales:

3. **The lattice BRST failure at $L = 8$, $\mathfrak{su}(3)$.** The mechanism (mixed second derivative) is structural and independent of L and the gauge algebra; numerical verification at full scale is deferred.
4. **The HCSM-04 IPR erratum.** The claimed orbit-dependent IPR ($2/128, 3/128, 4/128$) is not reproduced for plane waves. The uniform value is $1/64 = 2/128$ for all modes. This is flagged for correction.

XI. CONCLUSION

The substrate's BRST structure is $D = d + d^*$, not the imported continuum BRST operator. The physical state space is $\ker D$, of dimension 4, which is the emergent spacetime dimension. The bosonic and fermionic sectors of the Hodge complex have equal dimension 128, and D swaps them exactly; this is the substrate's native ghost cancellation, requiring no separate ghost fields. The naive lattice BRST fails in a specific, structural way — the mixed second derivative $L_1^x L_1^y (c^b c^c)$ — reflecting the lattice derivative not being a derivation of the Grassmann algebra. The diagonal flip F distinguishes B_1 from B_2 , making weak isospin selection a theorem. The partition function identity is $\mathcal{Z}_{\text{gf}} = \lim_{\beta \rightarrow \infty} \mathcal{Z}_{\text{sub}}(\beta) = \dim \ker D = 4$.

The framework's claim is that the Standard Model is what the substrate looks like when observed. Physics is what the machine cannot cancel. The Hodge complex is the machine. The kernel is what the machine cannot cancel.

ACKNOWLEDGMENTS

The author thanks the Entropia Research Institute for support.

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